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compact1, compact2, compact3

java.util.zip

Class CRC32

java.lang.Object
    java.util.zip.CRC32

All Implemented Interfaces:

Checksum

```
public class CRC32
extends Object
implements Checksum
```

A class that can be used to compute the CRC-32 of a data stream.

Passing a `null` argument to a method in this class will cause a `NullPointerException` to be thrown.

See Also:

Checksum

Constructor Summary

Constructors

Constructor	Description
`CRC32()`	Creates a new CRC32 object.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
long	**getValue**()	Returns CRC-32 value.
void	**reset**()	Resets CRC-32 to initial value.
void	**update**(byte[] b)	Updates the CRC-32 checksum with the specified array of bytes.
void	**update**(byte[] b, int off, int len)	Updates the CRC-32 checksum with the specified array of bytes.
void	**update**(**ByteBuffer** buffer)	Updates the checksum with the bytes from the specified buffer.
void	**update**(int b)	Updates the CRC-32 checksum with the specified byte (the low eight bits of the argument b).

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

CRC32

```
public CRC32()
```

Creates a new CRC32 object.

Method Detail

update

```
public void update(int b)
```

Updates the CRC-32 checksum with the specified byte (the low eight bits of the argument b).

Specified by:

`update` in interface `Checksum`

Parameters:

`b - the byte to update the checksum with`

update

```
public void update(byte[] b,
                   int off,
                   int len)
```

Updates the CRC-32 checksum with the specified array of bytes.

Specified by:

`update` in interface `Checksum`

Parameters:

`b - the byte array to update the checksum with`

`off - the start offset of the data`

`len - the number of bytes to use for the update`

Throws:

`ArrayIndexOutOfBoundsException - if off is negative, or len is negative, or off+len is greater than the length of the array b`

update

```
public void update(byte[] b)
```

Updates the CRC-32 checksum with the specified array of bytes.

Parameters:

`b - the array of bytes to update the checksum with`

update

```
public void update(ByteBuffer buffer)
```

Updates the checksum with the bytes from the specified buffer. The checksum is updated using buffer.`remaining()` bytes starting at buffer.`position()` Upon return, the buffer's position will be updated to its limit; its limit will not have been changed.

Parameters:

`buffer - the ByteBuffer to update the checksum with`

Since:

`1.8`

reset

```
public void reset()
```

Resets CRC-32 to initial value.

Specified by:

`reset` in interface `Checksum`

getValue

```
public long getValue()
```

Returns CRC-32 value.

Specified by:

`getValue` in interface `Checksum`

Returns:

`the current checksum value`

Submit a bug or feature

For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.

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